

2.8 setjmp.h

The setjmp header is used for controlling low-level calls and returns to and from functions.

Macros:

```
setjmp() ;
```

Functions:

```
longjmp() ;
```

Variables:

```
typedef jmp_buf
```

2.8.1 Variables and Definitions

The variable type `jmp_buf` is an array type used for holding information for `setjmp` and `longjmp`.

2.8.2 setjmp

Declaration:

```
int setjmp(jmp_buf environment) ;
```

Saves the environment into the variable *environment*. If a non-zero value is returned, then this indicates that the point in the sourcecode was reached by a `longjmp`. Otherwise zero is returned indicating the environment has been saved.

2.8.3 longjmp

Declaration:

```
void longjmp(jmp_buf environment, int value) ;
```

Causes the environment to be restored from a `setjmp` call where the environment variable had been saved. It causes execution to goto the `setjmp` location as if `setjmp` had returned the value of the variable *value*. The variable *value* cannot be zero. However, if zero is passed, then 1 is replaced. If the function where `setjmp` was called has terminated, then the results are undefined.

Example:

```
#include<setjmp.h>
#include<stdio.h>

void some_function(jmp_buf) ;
```

```
int main(void)
{
    int value;
    jmp_buf environment_buffer;

    value=setjmp(environment_buffer);
    if(value!=0)
    {
        printf("Reached this point from a longjmp with value=%d.\n",value);
        exit(0);
    }
    printf("Calling function.\n");
    some_function(environment_buffer);
    return 0;
}

void some_function(jmp_buf env_buf)
{
    longjmp(env_buf,5);
}
```

The output from this program should be:

Calling function.

Reached this point from a longjmp with value=5.